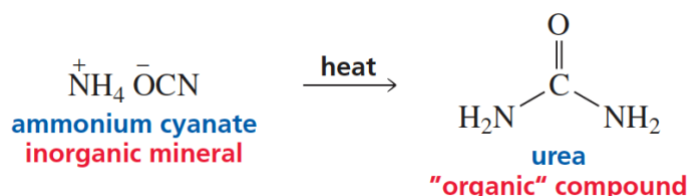


CML 101 Class Notes

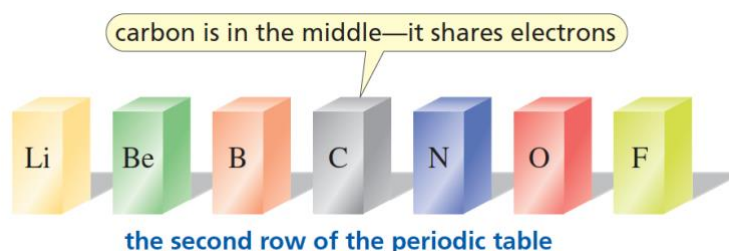
1. The Chemical Connection

To stay alive, early humans must have been able to distinguish between different kinds of materials in their world. “You can live on roots and berries,” they might have said, “but you can’t eat dirt. You can stay warm by burning tree branches, but you can’t burn rocks.” By the early eighteenth century, scientists thought they had grasped the nature of that difference, and in 1807, Jöns Jakob Berzelius gave names to the two kinds of materials. Compounds derived from living organisms were believed to contain an immeasurable vital force—the essence of life. These he called “organic.” Compounds derived from minerals—those lacking the vital force—were “inorganic.” Because chemists could not create life in the laboratory, they assumed they could not create compounds that have a vital force. You can imagine their surprise when, in 1828, Friedrich Wöhler produced urea—a compound excreted by mammals—by heating ammonium cyanate, an inorganic mineral.



For the first time, an “organic” compound had been obtained from something other than a living organism and, therefore, did not contain a vital force. Thus, chemists needed a new definition for “organic compounds.” Organic compounds are now defined as compounds that are based on carbon.

Why are there so many carbon-containing compounds? The answer lies in carbon’s position in the periodic table. Carbon is in the center of the second row of elements. We will see that the atoms to the left of carbon have a tendency to give up electrons, whereas the atoms to the right have a tendency to accept electrons.



Because carbon is in the middle, it neither readily gives up nor readily accepts electrons. Instead, it shares electrons. Carbon can share electrons with several kinds of atoms as well as with other carbon atoms. Consequently, carbon forms millions of stable compounds with a wide range of chemical properties simply by sharing electrons.

When we study organic chemistry, we learn how organic compounds react. Organic compounds consist of atoms held together by covalent bonds. When an organic compound reacts, some of these covalent bonds break and some new covalent bonds form.

Covalent bonds form when two atoms share electrons, and they break when two atoms no longer share electrons.

How easily a covalent bond forms or breaks depends on the electrons that are shared, which, in turn, depends on the atoms to which the electrons belong.

2. Natural Versus Synthetic Organic Compounds

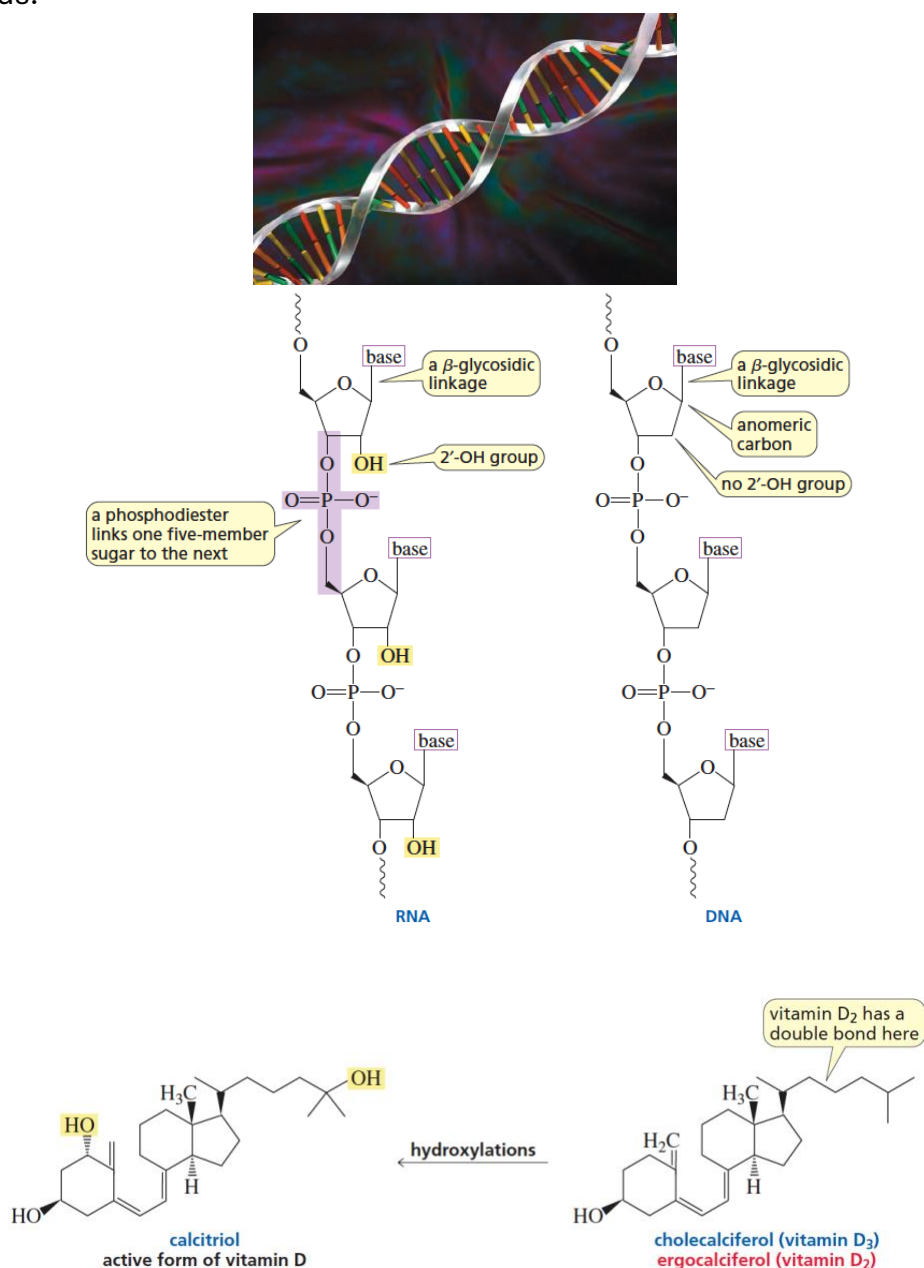
It is a popular belief that natural substances—those made in nature—are superior to synthetic ones—those made in the laboratory. Yet when a chemist synthesizes a compound, such as penicillin or morphine, the compound is the same in all respects as the compound synthesized in nature. Sometimes chemists can even improve on nature. For example, chemists have synthesized analogues of penicillin—compounds with structures similar to that of penicillin—that do not produce the allergic responses that a significant fraction of the population experiences from naturally produced penicillin or that do not have the bacterial resistance of the naturally produced antibiotic. Chemists have also synthesized analogues of morphine that have the same pain-killing effects but, unlike morphine, are not habit-forming. Most commercial morphine is obtained from opium, the juice extracted from the species of poppy shown in the photo. Morphine is the starting material for the synthesis of heroin. One of the side products formed in the synthesis has an extremely pungent odour; dogs used by drug enforcement agencies are trained to recognize this odour. Nearly three-quarters of the world's supply of heroin comes from the poppy fields of Afghanistan.



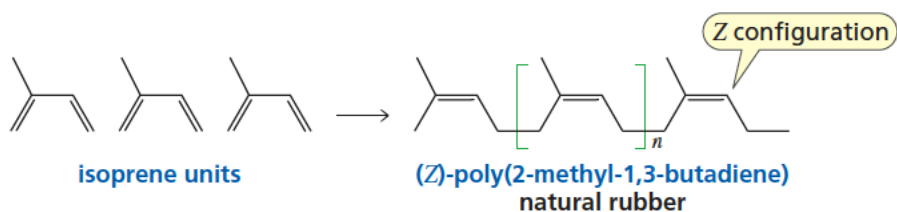
a field of poppies in Afghanistan

Why is an entire branch of chemistry devoted to the study of carbon-containing compounds? We study organic chemistry because just about all of the compounds that make life possible and that make us who we are—proteins, enzymes, vitamins, lipids, carbohydrates, DNA, RNA—are organic compounds. Thus, the chemical reactions that

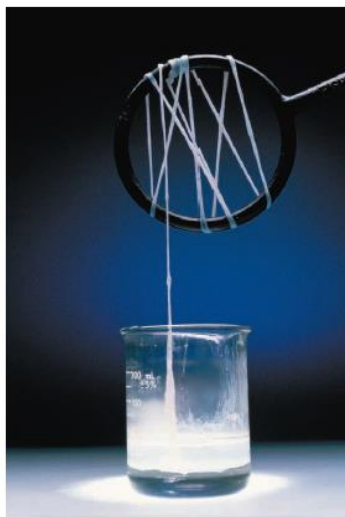
take place in living systems, including our own bodies, are reactions of organic compounds.



Most of the compounds found in nature—those that we rely on for food, clothing (cotton, wool, silk), and energy (natural gas, petroleum)—are organic compounds as well. Natural rubber is a polymer of 2-methyl-1,3-butadiene, also called isoprene. On average a molecule of rubber contains 5000 isoprene units.



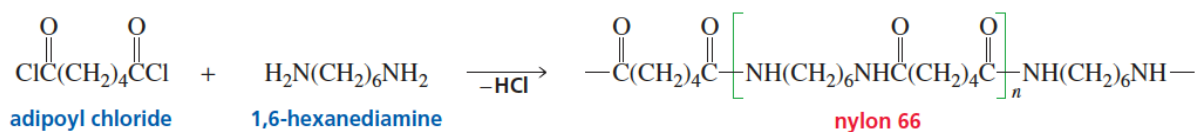
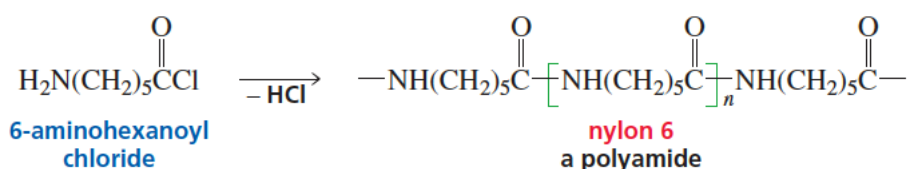
Some synthetic organic compounds prevent shortages of naturally occurring compounds. For example, it has been estimated that if synthetic materials—nylon, polyester, Lycra—were not available for clothing, all of the arable land in the United States would have to be used for the production of cotton and wool just to provide enough material to clothe us. Synthetic polymers are synthesized by scientists. Natural polymers such as cotton, wool and synthetic polymers such as nylon, polyester are used for clothing.



Nylon is pulled from a beaker of adipoyl chloride and 1,6-hexanediamine.

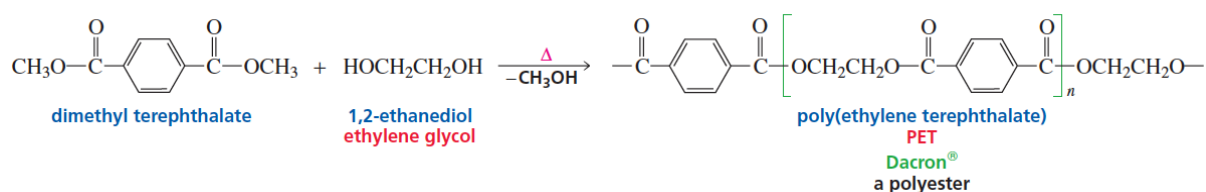


nylon rope





Mylar balloons

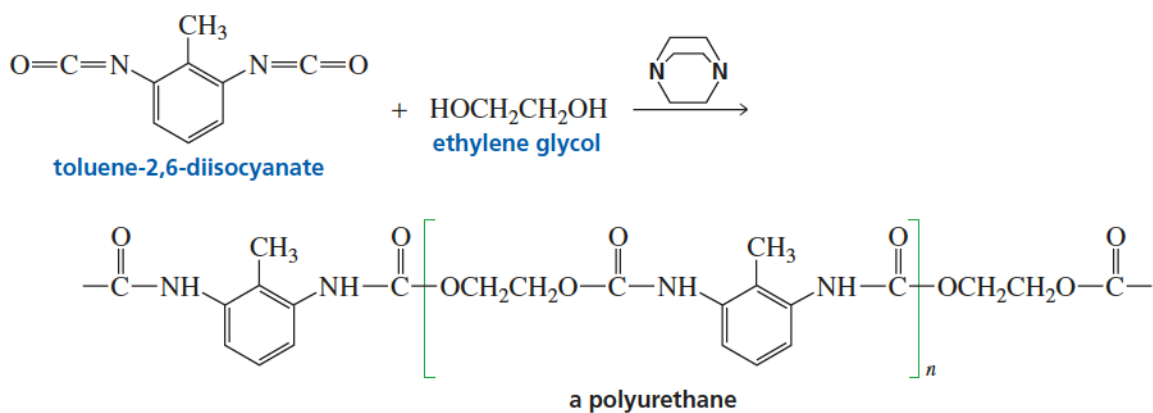


Health concern: Phthalates

Phthalates have been found to be endocrine disruptors—that is, they can alter the proper balance of hormones. Therefore, the primary risk they pose is to a developing fetus. It is difficult to avoid phthalates because of the numerous items (for example, the linings of most aluminum food and beverage cans) that contain them.



polyurethane foam

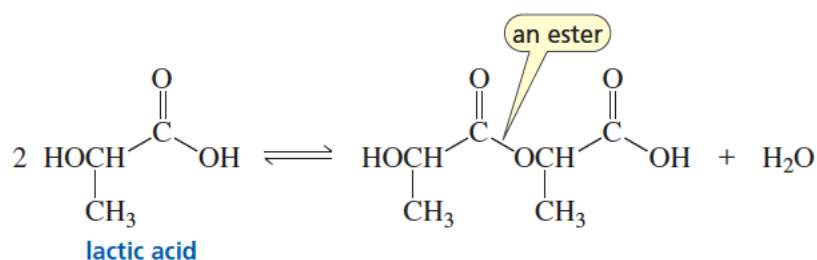


One of the most important uses of polyurethanes is in fabrics with elastic properties, such as Lycra-known generically as spandex.

Biodegradable polymers are polymers that can be degraded by microorganisms such as bacteria, fungi, or algae. Polylactide (PLA), a biodegradable polymer of lactic acid, has found wide use. When lactic acid is polymerized, a molecule of water is lost that can hydrolyze the new ester bond, reforming lactic acid. Because lactic acid has an asymmetric center, there are several different forms of the polymer. The polymer's physical properties depend on the ratio of R and S enantiomers used in its synthesis. Polylactides are currently being used in nonwrinkle fabrics, microwavable trays, food packaging, and in several medical applications such as sutures, stents, and drug-delivery devices. They are also used for cold drink glasses. Unfortunately, hot drinks cause the polymer to liquify.



glasses made of PLA



Recycling polymers

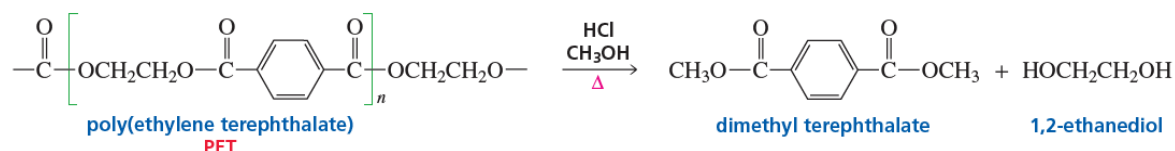


compressed plastic bottles waiting to be recycled

Only polymers with the two lowest numbers PET (1)—the polymer used to make soft drink bottles—and HDPE (2)—the denser polymer used for juice and milk bottles—are

recycled to any significant extent. This amounts to less than 25% of all polymers. The others are found in landfills.

PET is recycled by heating the polymer in an acidic solution of methanol. This transesterification reaction is the reverse of the transesterification reaction that formed the polymer. Because the products of recycling PET are the monomers used to make it, the products can be used to make more PET.

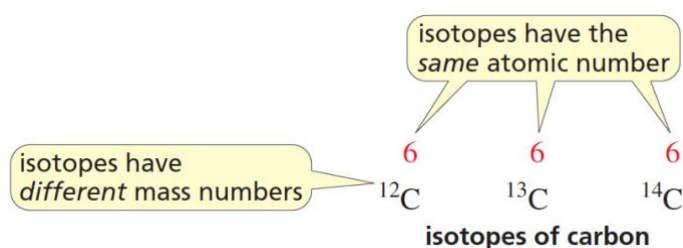


Melamine poisoning

A few years ago, Chinese milk products were found to have been deliberately contaminated with melamine. Although melamine was never approved as a food additive, milk dealers added it to diluted milk to increase its apparent protein concentration. This milk was then used to make such things as powdered milk, infant formulas, cookies, and other foods. The contaminated milk sickened 300,000 Chinese children and killed at least 6 babies. The previous year, about 1000 dogs and cats died in the United States as a result of eating pet food with melamine-containing ingredients traced to Chinese suppliers. As a result of these tragedies, the Chinese legislature passed a food safety law and now requires producers to list all additives on food labels.

3. Isotopes

The atomic number of an atom is the number of protons in its nucleus. The atomic number is unique to a particular element. For example, the atomic number of carbon is 6, which means that all uncharged carbon atoms have six protons and six electrons. Although atoms can gain electrons and become negatively charged or lose electrons and become positively charged, the number of protons in an atom of a particular element never changes. The mass number of an atom is the sum of its protons and neutrons. Although all carbon atoms have the same atomic number, they do not all have the same mass number. Why? Because carbon atoms can have varying numbers of neutrons. For example, 98.89% of all carbon atoms have six neutrons—giving them a mass number of 12—and 1.11% have seven neutrons—giving them a mass number of 13. These two different kinds of carbon atoms (^{12}C and ^{13}C) are called isotopes.



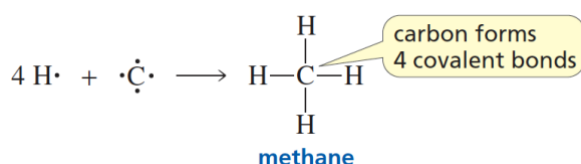
Carbon also contains a trace amount of ^{14}C , which has six protons and eight neutrons. This isotope of carbon is radioactive, decaying with a half-life of 5730 years (The half-life

is the time it takes for one-half of the nuclei to decay). As long as a plant or an animal is alive, the ^{14}C that is lost through exhalation or excretion is constantly replenished. When it dies, however, it no longer ingests ^{14}C . Consequently, its ^{14}C is slowly lost through radioactive decay. Therefore, the age of a substance derived from a living organism can be determined by its ^{14}C content.

The atomic mass is the weighted average of the isotopes in the element. Because an atomic mass unit (amu) is defined as exactly $1/12$ of the mass of ^{12}C , the mass of ^{12}C is 12.0000 amu; the mass of ^{13}C is 13.0035 amu. Therefore, the atomic mass of carbon is 12.011 amu because $(0.9889 \times 12.0000) + (0.0111 \times 13.0035) = 12.011$. The molecular mass is the sum of the atomic masses of all the atoms in the molecule.

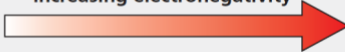
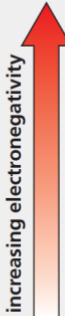
4. Nonpolar and Polar Covalent Bonds

Covalent bonds are classified as nonpolar or polar depending on the difference in the electronegativities of the atoms that share the electrons.



Electronegativity is a measure of the ability of an atom to pull electrons toward itself. The electronegativities of some of the elements are shown in Table below. Notice that electronegativity increases from left to right across a row of the periodic table and from bottom to top in any of the columns.

1A	2A		3A	4A	5A	6A	7A
H 2.1							
Li 1.0	Be 1.5		B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
Na 0.9	Mg 1.2		Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0
K 0.8	Ca 1.0						Br 2.8
							I 2.5

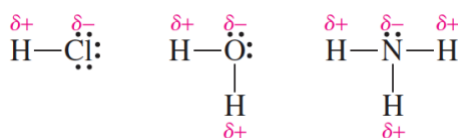
^aElectronegativity values are relative, not absolute. As a result, there are several scales of electronegativities. The electronegativities listed here are from the scale devised by Linus Pauling.

If the electronegativity difference between the bonded atoms is less than 0.5, then the bond is a nonpolar covalent bond. That is, the atoms share the bonding electrons equally—the electrons represented by the bond are symmetrically distributed around each atom. Examples of nonpolar covalent bonds are shown below.

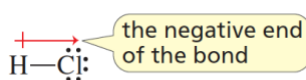


If the electronegativity difference between the bonded atoms is between 0.5 and about 1.9, then the bond is a polar covalent bond. The bonding electrons are unsymmetrically

distributed because the bonding atoms have sufficiently different electronegativities; one end of a polar covalent bond has a partial negative charge (δ^-), and one end has a partial positive charge (δ^+). The negative end of the bond is the end with the more electronegative atom. The greater the difference in electronegativity between the bonded atoms, the more polar the bond.

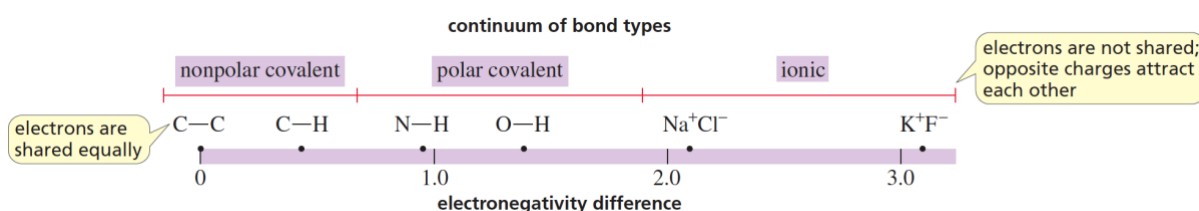


The direction of bond polarity can be indicated with an arrow. By convention, chemists draw the arrow in the direction the electrons are pulled. Thus, the head of the arrow is at the negative end of the bond; a short perpendicular line near the tail of the arrow marks the positive end of the bond.



If the electronegativity difference between the atoms is greater than 1.9, the atoms do not share their electrons. One of the atoms transfers an electron to the other, and the resulting ions (for example, Na^+ and Cl^-) are held together by electrostatic attraction—an attractive force between opposite charges. Sodium chloride is an example of an ionic compound (also known as a salt). Ionic compounds are formed when an element on the left side of the periodic table transfers one or more electrons to an element on the right side of the periodic table.

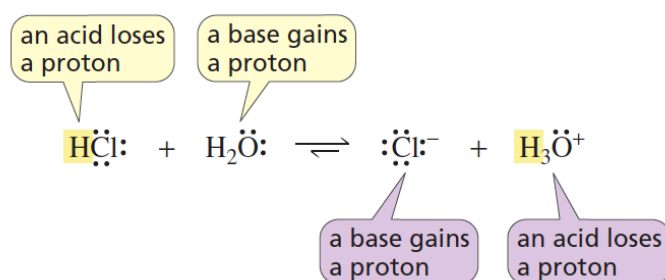
You can think of nonpolar covalent bonds—where the electrons are shared equally—being at one end of the continuum of bond types and ionic compounds—where no electrons are shared—at the other end. Polar covalent bonds fall somewhere in between.



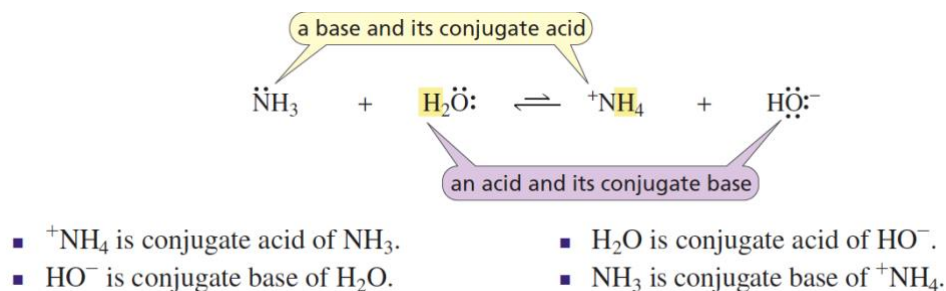
5. Acids and Bases

The chemistry of acids and bases explains things such as the cause of acid rain and why it destroys monuments and plants, why exercise increases the rate of breathing, why blood has to be buffered and how that buffering is accomplished. Acids and bases play an important role in organic chemistry.

There are two definitions for the terms acid and base, the Brønsted–Lowry definitions and the Lewis definitions. According to Brønsted and Lowry, an acid is a species that loses a proton, and a base is a species that gains a proton. For example, in the reaction shown below, hydrogen chloride (HCl) is an acid because it loses a proton, and water is a base because it gains a proton. The reaction of an acid with a base is called an acid–base reaction or a proton transfer reaction. Notice that the reverse of an acid–base reaction is also an acid–base reaction. In the reverse reaction, H_3O^+ is an acid because it loses a proton, and Cl^- is a base because it gains a proton.



Another example of an acid–base reaction is the reaction between ammonia and water: ammonia (NH_3) is a base because it gains a proton, and water is an acid because it loses a proton. In the reverse reaction, ammonium ion ($^+\text{NH}_4$) is an acid because it loses a proton, and hydroxide ion (HO^-) is a base because it gains a proton.

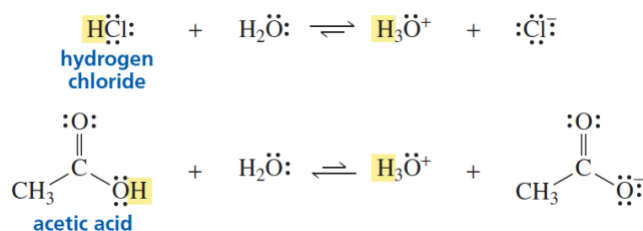


Acidity is a measure of the tendency of a compound to lose a proton, whereas **basicity** is a measure of a compound's affinity for a proton. A strong acid has a strong tendency to lose a proton. This means that its conjugate base must be weak because it has little affinity for the proton. A weak acid has little tendency to lose its proton, indicating that its conjugate base is strong because it has a high affinity for the proton. Thus, the following important relationship exists between an acid and its conjugate base:

The stronger the acid, the weaker its conjugate base.

pK_a and pH

When a strong acid such as hydrogen chloride is dissolved in water, almost all the molecules dissociate (break into ions), which means that the products are favored at equilibrium—the equilibrium lies to the right. When a much weaker acid, such as acetic acid, is dissolved in water, very few molecules dissociate, so the reactants are favored at equilibrium—the equilibrium lies to the left. A longer arrow is drawn toward the species favored at equilibrium.



K_q

The degree to which an acid (HA) dissociates in an aqueous solution is indicated by the equilibrium constant of the reaction, K_{eq} . Brackets are used to indicate the concentrations of the reactants and products (in moles/liter).



$$K_{\text{eq}} = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}][\text{H}_2\text{O}]}$$

K_a

The degree to which an acid (HA) dissociates is normally determined in a dilute solution, so the concentration of water remains essentially constant. Combining the two constants (K_{eq} and $[\text{H}_2\text{O}]$) allows the equilibrium expression to be rewritten using a new equilibrium constant, K_{a} , called the acid dissociation constant.

$$K_{\text{a}} = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]} = K_{\text{eq}} [\text{H}_2\text{O}]$$

pK_a

The larger the acid dissociation constant, the stronger the acid—that is, the greater its tendency to lose a proton. Hydrogen chloride, with an acid dissociation constant of 107, is a stronger acid than acetic acid, with an acid dissociation constant of 1.74×10^{-5} . For convenience, the strength of an acid is generally indicated by its pK_{a} value rather than its K_{a} value, where

$$\text{pK}_{\text{a}} = -\log K_{\text{a}}$$

The pK_{a} of hydrogen chloride is -7, and the pK_{a} of acetic acid, a much weaker acid, is 4.76. Notice that the stronger the acid, the smaller its pK_{a} value.

very strong acids	$\text{pK}_{\text{a}} < 1$
moderately strong acids	$\text{pK}_{\text{a}} = 1-3$
weak acids	$\text{pK}_{\text{a}} = 3-5$
very weak acids	$\text{pK}_{\text{a}} = 5-15$
extremely weak acids	$\text{pK}_{\text{a}} > 15$

pH

The concentration of protons in a solution is indicated by pH. This concentration is written as either $[H^+]$ or, because a proton in water is solvated, as $[H_3O]^+$.

$$pH = -\log [H^+]$$

Solution	pH
NaOH, 1.0 M	14
NaOH, 0.1 M	13
Household bleach	12
Household ammonia	11
Milk of magnesia	10
Borax	9
Baking soda	8
Egg white, seawater	8
Human blood, tears	7
Milk	6
Saliva	6
Rain	5
Coffee	5
Tomatoes	4
Wine	4
Cola, vinegar	3
Lemon juice	2
Gastric juice	1
HCl, 0.1 M	1
HCl, 1.0 M	0

Because pH values decrease as the acidity of the solution increases, we see that lemon juice is more acidic than coffee and that rain is more acidic than milk. Solutions with pH values less than 7 are acidic, whereas those with pH values greater than 7 are basic. The pH of a solution can be changed simply by adding acid or base to the solution. Do not confuse pH and pKa. The pH scale is used to describe the acidity of a solution, whereas the pKa indicates the tendency of a compound to lose its proton. Thus, the pKa is characteristic of a particular compound, much like a melting point or a boiling point.

Acid Rain

Rain is mildly acidic ($pH = 5.5$) because water reacts with the CO_2 in the air to form carbonic acid (a weak acid with a pK_a value of 6.4).



In some parts of the world, rain has been found to be much more acidic (pH values as low as 4.3). This so-called acid rain is formed where sulfur dioxide and nitrogen oxides are produced, because water reacts with these gases to form strong acids—sulfuric acid ($pK_a = -5.0$) and nitric acid ($pK_a = -1.3$). Burning fossil fuels for the generation of electric power is the factor most responsible for forming these acid-producing gases.

Acid rain has many deleterious effects. It can destroy aquatic life in lakes and streams; it can make soil so acidic that crops cannot grow and forests are destroyed (see p. 50); and it can cause the deterioration of paint and building materials, including monuments and statues that are part of our cultural heritage. Marble—a form of calcium carbonate—decays because protons react with CO_3^{2-} in the marble to form carbonic acid, which decomposes to CO_2 and H_2O (the reverse of the reaction shown above on the left).

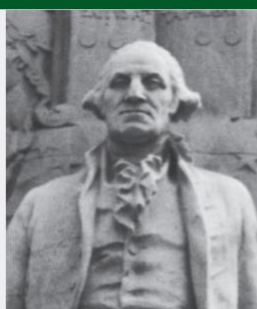
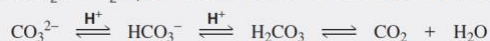


photo taken in 1935

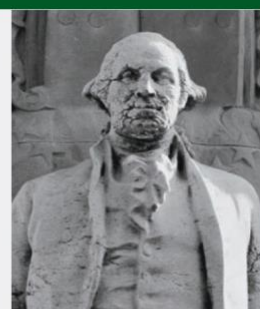


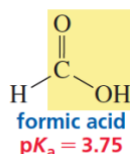
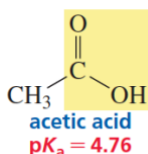
photo taken in 1994

Statue of George Washington in Washington Square Park in Greenwich Village, New York.

Organic Acids and Bases

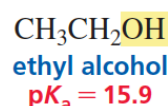
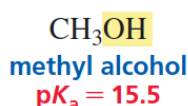
Carboxylic Acids

The most common organic acids are carboxylic acids—compounds that have a COOH group. Acetic acid and formic acid are examples of carboxylic acids. Carboxylic acids have pKa values ranging from about 3 to 5, so they are weak acids.



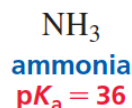
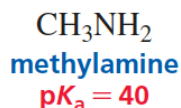
Alcohols

Alcohols—compounds that have an OH group—are much weaker acids than carboxylic acids, with pKa values close to 16. Methyl alcohol and ethyl alcohol are examples of alcohols.



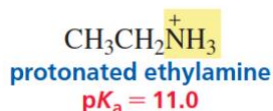
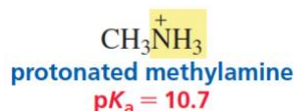
Amines

Amines are compounds that result from replacing one or more of the hydrogens bonded to ammonia with a carbon-containing substituent. Amines and ammonia have such high pKa values that they rarely behave as acids—they are more likely to act as bases. In fact, they are the most common organic bases.

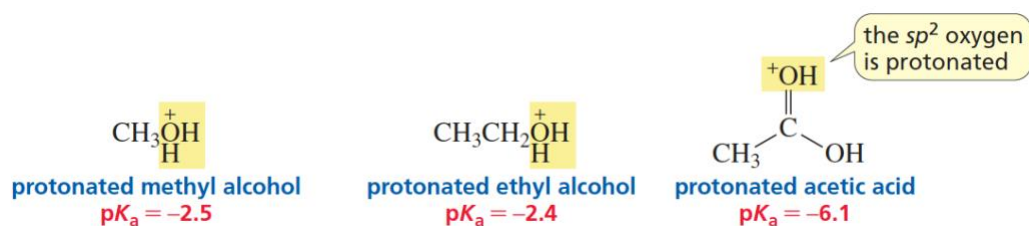


Protonated Compounds

We can assess the strength of a base by considering the strength of its conjugate acid—remembering that the stronger the acid, the weaker its conjugate base. For example, based on their pKa values, protonated methylamine (10.7) is a stronger acid than protonated ethylamine (11.0), which means that methylamine is a weaker base than ethylamine. pKa values of protonated amines are about 11.



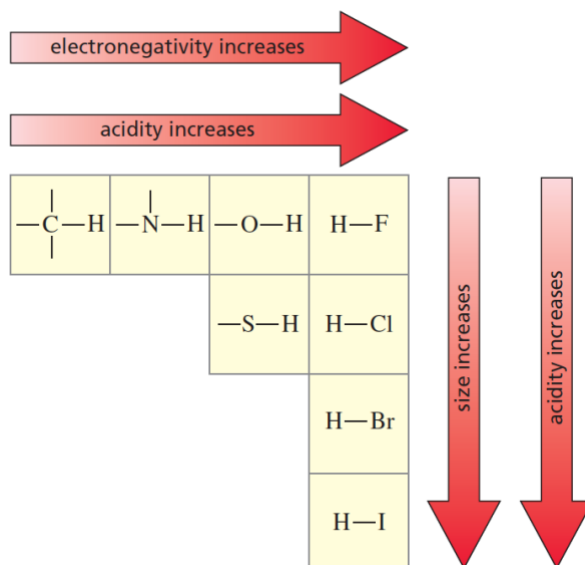
Protonated alcohols and protonated carboxylic acids are very strong acids, with pKa values 6 0.



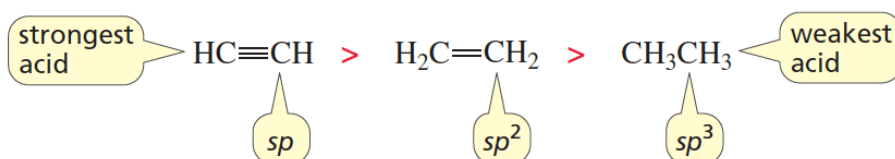
Factors that determine acid strength

The strength of an acid depends on five factors: the size of the atom to which the hydrogen is attached, the electronegativity of the atom to which the hydrogen is attached, the hybridization of the atom to which the hydrogen is attached, inductive electron withdrawal, and electron delocalization. All five factors affect acidity by affecting the stability of the conjugate base.

1. Size: As the atom attached to the hydrogen increases in size (going down a column of the periodic table), the strength of the acid increases.
2. Electronegativity: As the atom attached to the hydrogen increases in electronegativity.



3. Hybridization: The electronegativity of an atom changes with hybridization as follows: $sp > sp^2 > sp^3$. Because an sp carbon is the most electronegative, a hydrogen attached to an sp carbon is the most acidic, and a hydrogen attached to an sp^3 carbon is the least acidic.



4. Inductive electron withdrawal: An electron-withdrawing group increases the strength of

an acid. As the electronegativity of the electron-withdrawing group increases or as it moves closer to the acidic hydrogen, the strength of the acid increases.

pK _a Values					
Compound	pK _a	Compound	pK _a	Compound	pK _a
<chem>CH3C#N[H+]</chem>	-10.1	<chem>O=[N+]([O-])c1ccc(N)cc1</chem>	1.0	<chem>CC(=O)Oc1ccc(C)cc1</chem>	4.3
<chem>HI</chem>	-10	<chem>c1ccncc1</chem>	1.0	<chem>COc1ccc(C(=O)O)cc1</chem>	4.5
<chem>HBr</chem>	-9	<chem>[NH+]1CCCCN1</chem>	1.2	<chem>c1ccccc1[NH3+]</chem>	4.6
<chem>CH3C(=O)[OH+]</chem>	-8	<chem>O=Cc1ccc(N)cc1</chem>	1.3	<chem>CC(=O)O</chem>	4.8
<chem>CH3C(=O)C(=O)[OH+]</chem>	-7.3	<chem>ClCC(=O)O</chem>	1.9	<chem>c1ccc2c(c1)cnc2</chem>	4.9
<chem>HCl</chem>	-7	<chem>[H3PO4]</chem>	2.0	<chem>CC(=O)O</chem>	5.1
<chem>c1ccccc1S(=O)(=O)[H]</chem>	-6.5	<chem>[HSO4-]</chem>	2.5	<chem>c1ccccc1[NH3+]</chem>	5.2
<chem>CH3C(=O)C(=O)[OH+]</chem>	-6.5	<chem>c1cnc2c(c1)cnc2</chem>	2.7	<chem>COc1ccc(N)cc1</chem>	5.3
<chem>CH3C(=O)C(=O)[OH+]</chem>	-6.1	<chem>FCC(=O)O</chem>	2.8	<chem>CC(C)([NH3+])</chem>	5.5
<chem>H2SO4</chem>	-5	<chem>ClCC(=O)O</chem>	2.9	<chem>CC(C)(=O)C(=O)O</chem>	5.9
<chem>c1cc[nH+]c1</chem>	-3.8	<chem>BrCC(=O)O</chem>	3.2	<chem>O=[NH3+]</chem>	6.0
<chem>CH3CH2C(=O)O</chem>	-3.6	<chem>ICH2C(=O)O</chem>	3.2	<chem>H2CO3</chem>	6.4
<chem>CH3CH2C(=O)O</chem>	-2.4	<chem>HF</chem>	3.2	<chem>[H2PO4-]</chem>	6.7
<chem>CH3C(=O)O</chem>	-2.5	<chem>HNO2</chem>	3.4	<chem>c1cc[nH+]c1</chem>	6.8
<chem>H3O+</chem>	-1.7	<chem>O=[O]N</chem>	3.4	<chem>H2S</chem>	7.0
<chem>HNO3</chem>	-1.3	<chem>O=[O]N</chem>	3.4	<chem>O=[O]Nc1ccc(O)cc1</chem>	7.1
<chem>CH3SO3H</chem>	-1.2	<chem>CC(=O)c1ccc(C(=O)O)cc1</chem>	3.7	<chem>O=Cc1ccc(O)cc1</chem>	7.7
<chem>CH3C(=O)NH2</chem>	0.0	<chem>O=Cc1ccc(C(=O)O)cc1</chem>	3.8	<chem>c1ccccc1S</chem>	7.8
<chem>F3CC(=O)O</chem>	0.2	<chem>BrCc1ccc(N)cc1</chem>	3.9		
<chem>Cl3CC(=O)O</chem>	0.64	<chem>BrCc1ccc(C(=O)O)cc1</chem>	4.0		
<chem>c1cc[nH+]c1</chem>	0.79	<chem>c1ccccc1C(=O)O</chem>	4.2		

^apK_a values are for the red H in each structure

pK_a Values (continued)

Compound	pK _a	Compound	pK _a	Compound	pK _a
	8.0		10.7		~17
	8.1		10.7		17
	8.2		10.7		18
	8.3		10.7		20
	8.6		11.1		24.5
	8.9		11.0		25
	9.1		11.3		25
	9.3		11.6		30
	9.4		12.4		35
	9.4		12.4		36
	9.5		13.3		36
	9.8		13.5		40
	10.0		13.9		41
	10.1		14.4		43
	10.2		15.5		43
	10.2		15.7		44
	10.2		16.0		46
	10.3		16		60
	10.5		16.0		> 60
	10.6				